In [1]:
```
pip install seaborn
```

```
Requirement already satisfied: seaborn in /Users/damirdarasu/Documents/AI&
DS/.venv/lib/python3.13/site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in /Users/damirdarasu/
Documents/AI&DS/.venv/lib/python3.13/site-packages (from seaborn) (2.3.2)
Requirement already satisfied: pandas>=1.2 in /Users/damirdarasu/Document
s/AI&DS/.venv/lib/python3.13/site-packages (from seaborn) (2.3.2)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in /Users/damirdara
su/Documents/AI&DS/.venv/lib/python3.13/site-packages (from seaborn) (3.1
0.6)
Requirement already satisfied: contourpy>=1.0.1 in /Users/damirdarasu/Docu
ments/AI&DS/.venv/lib/python3.13/site-packages (from matplotlib!=3.6.1,>=
3.4->seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /Users/damirdarasu/Document
s/AI&DS/.venv/lib/python3.13/site-packages (from matplotlib!=3.6.1,>=3.4->
seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /Users/damirdarasu/Doc
uments/AI&DS/.venv/lib/python3.13/site-packages (from matplotlib!=3.6.1,>=
3.4->seaborn) (4.59.2)
Requirement already satisfied: kiwisolver>=1.3.1 in /Users/damirdarasu/Doc
uments/AI&DS/.venv/lib/python3.13/site-packages (from matplotlib!=3.6.1,>=
3.4->seaborn) (1.4.9)
Requirement already satisfied: packaging>=20.0 in /Users/damirdarasu/Docum
ents/AI&DS/.venv/lib/python3.13/site-packages (from matplotlib!=3.6.1,>=3.
4->seaborn) (25.0)
Requirement already satisfied: pillow>=8 in /Users/damirdarasu/Documents/A
I&DS/.venv/lib/python3.13/site-packages (from matplotlib!=3.6.1,>=3.4->sea
born) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /Users/damirdarasu/Docu
ments/AI&DS/.venv/lib/python3.13/site-packages (from matplotlib!=3.6.1,>=
3.4->seaborn) (3.2.3)
Requirement already satisfied: python-dateutil>=2.7 in /Users/damirdarasu/
Documents/AI&DS/.venv/lib/python3.13/site-packages (from matplotlib!=3.6.
1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /Users/damirdarasu/Document
s/AI&DS/.venv/lib/python3.13/site-packages (from pandas>=1.2->seaborn) (20
25.2)
Requirement already satisfied: tzdata>=2022.7 in /Users/damirdarasu/Docume
nts/AI&DS/.venv/lib/python3.13/site-packages (from pandas>=1.2->seaborn)
(2025.2)
Requirement already satisfied: six>=1.5 in /Users/damirdarasu/Documents/AI
&DS/.venv/lib/python3.13/site-packages (from python-dateutil>=2.7->matplot
lib!=3.6.1,>=3.4->seaborn) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

In [3]:
```
import pandas as pd
df_titanic = pd.read_csv("titanic.csv")
df_titanic.head()
```

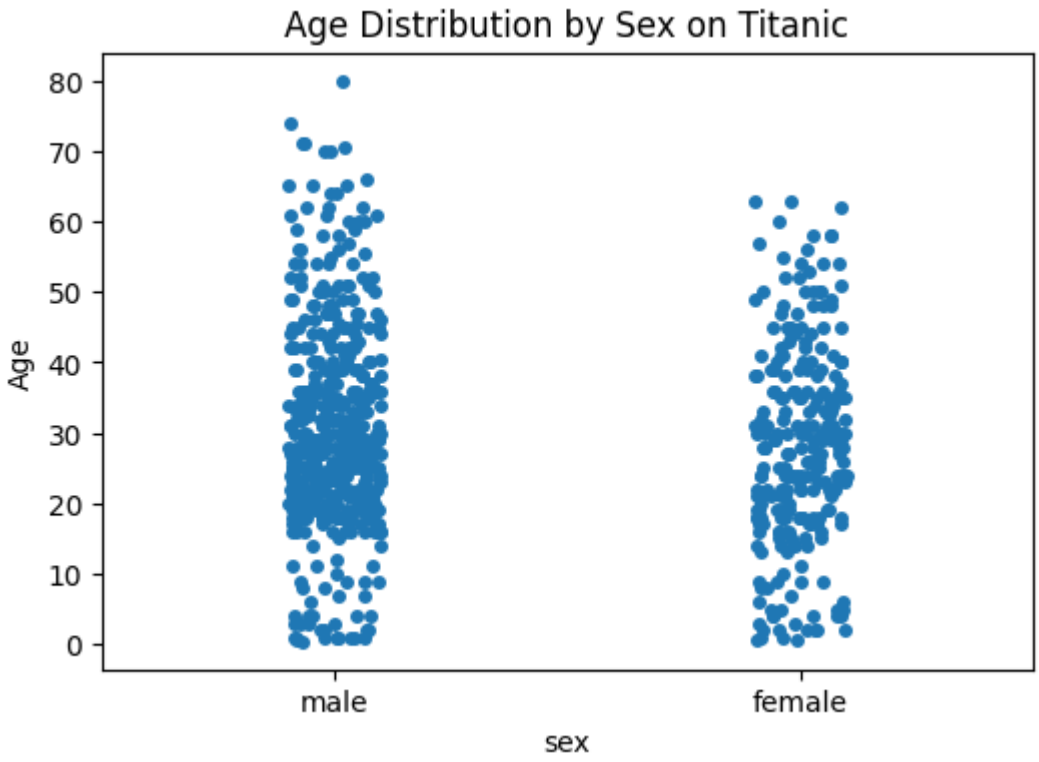
Out[3]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450

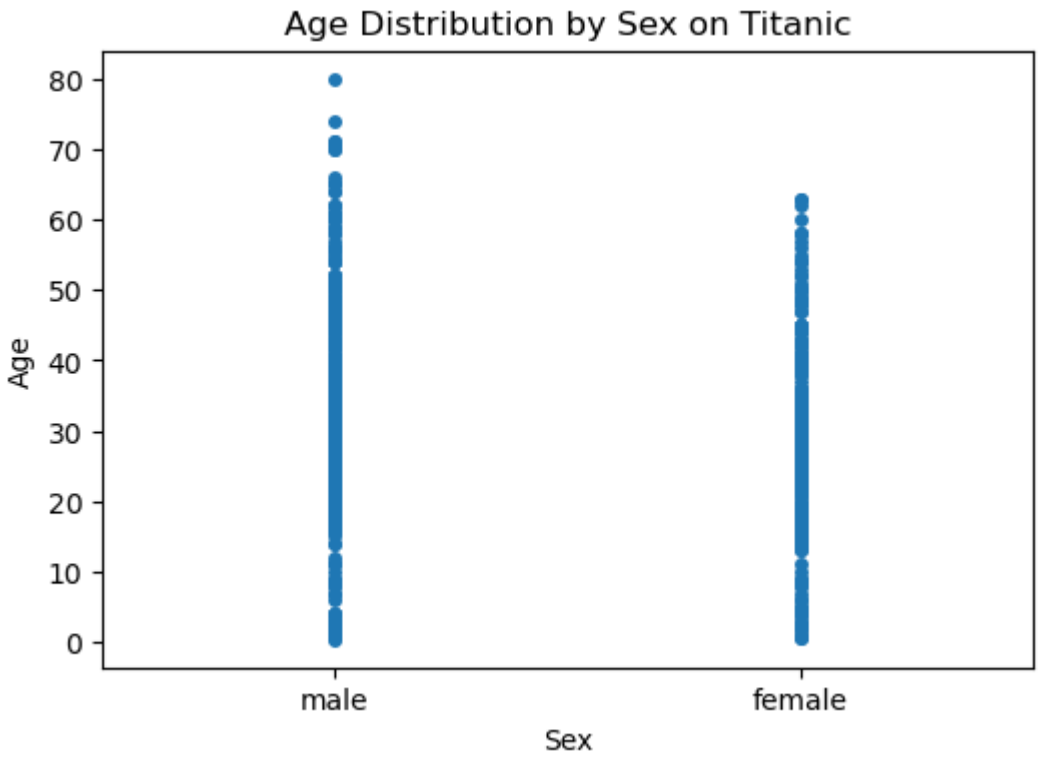
Strip plot

In [4]:

```
import seaborn as sns
import matplotlib.pyplot as plt
titanic = sns.load_dataset('titanic')
plt.figure(figsize=(6, 4))
sns.stripplot(x='sex', y='age', data=titanic, jitter=True)
plt.title('Age Distribution by Sex on Titanic')
plt.xlabel('sex')
plt.ylabel('Age')
plt.show()
```

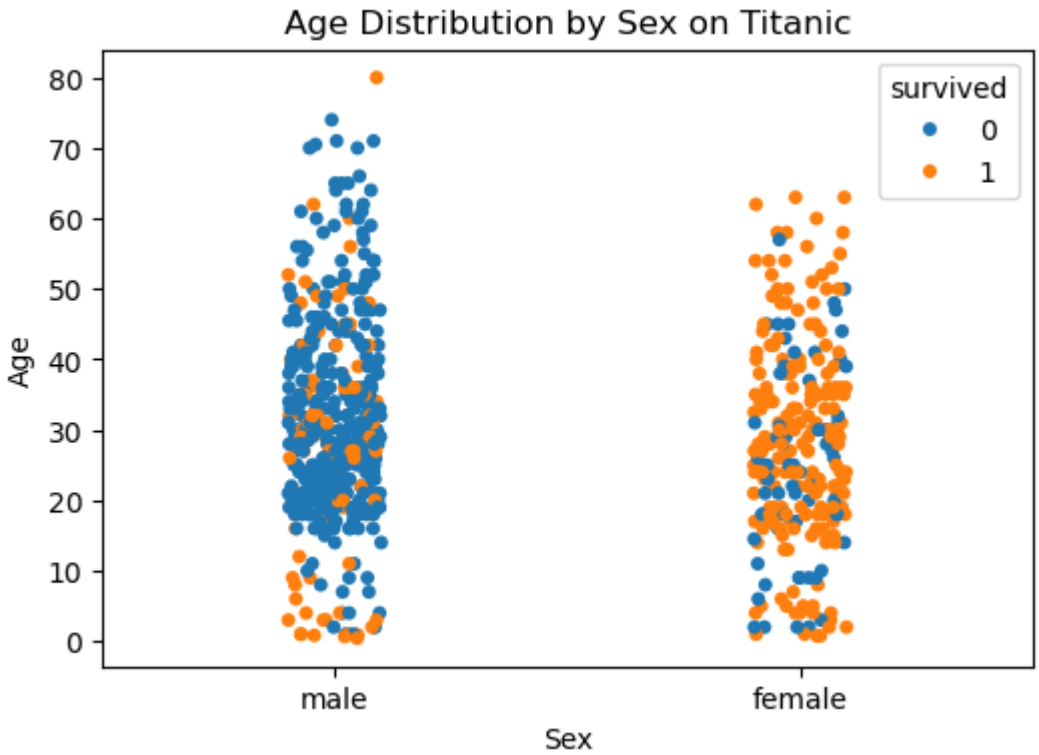


In [4]:
```
import seaborn as sns
import matplotlib.pyplot as plt
titanic = sns.load_dataset('titanic')
plt.figure(figsize=(6, 4))
sns.stripplot(x='sex', y='age', data=titanic, jitter=False)
plt.title('Age Distribution by Sex on Titanic')
plt.xlabel('Sex')
plt.ylabel('Age')
plt.show()
```

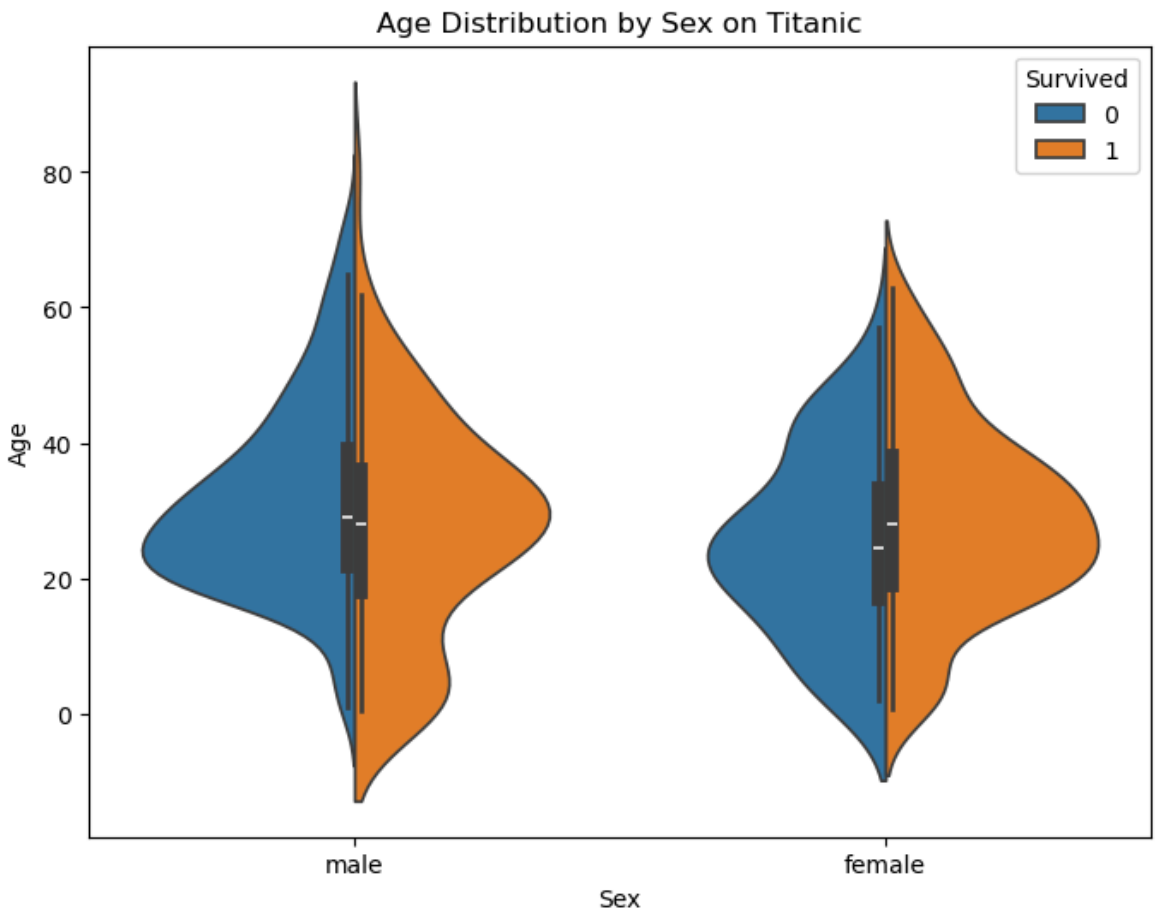


In [5]:
```
import seaborn as sns
import matplotlib.pyplot as plt
```

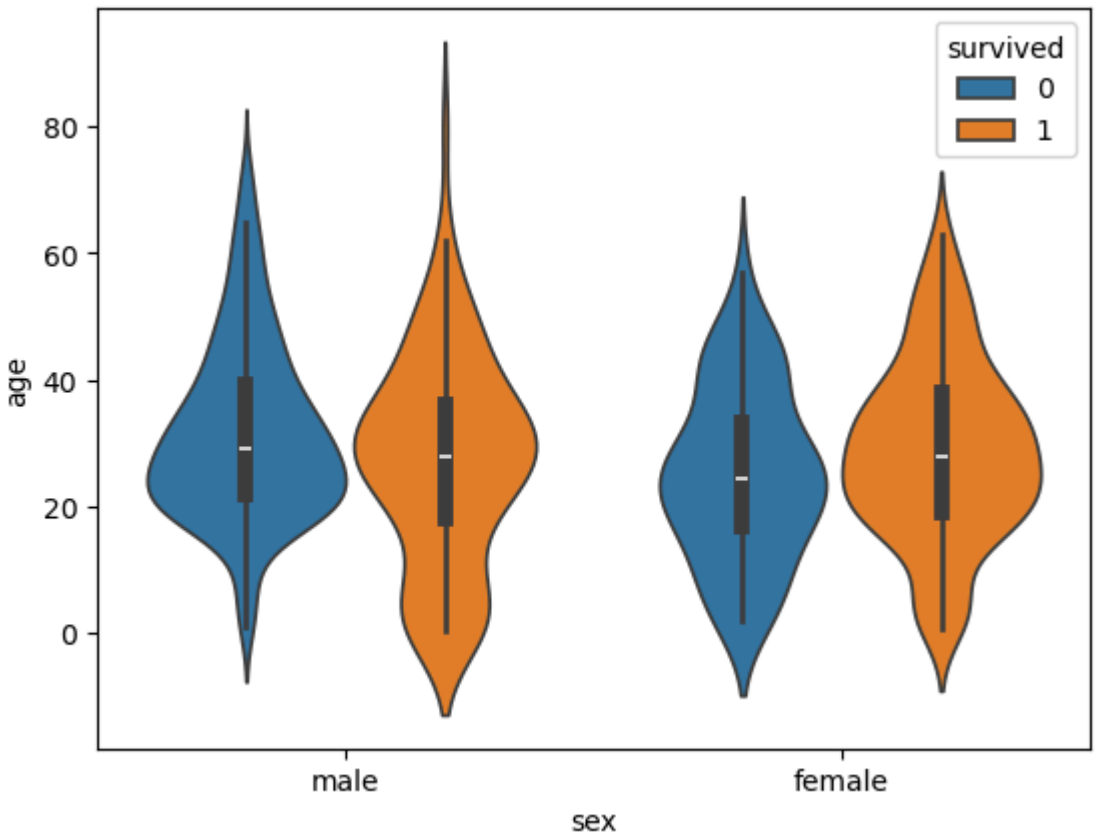
```
titanic = sns.load_dataset('titanic')
plt.figure(figsize=(6, 4))
sns.stripplot(x='sex', y='age', hue='survived', data=titanic, jitter=True
plt.title('Age Distribution by Sex on Titanic')
plt.xlabel('Sex')
plt.ylabel('Age')
plt.show()
```



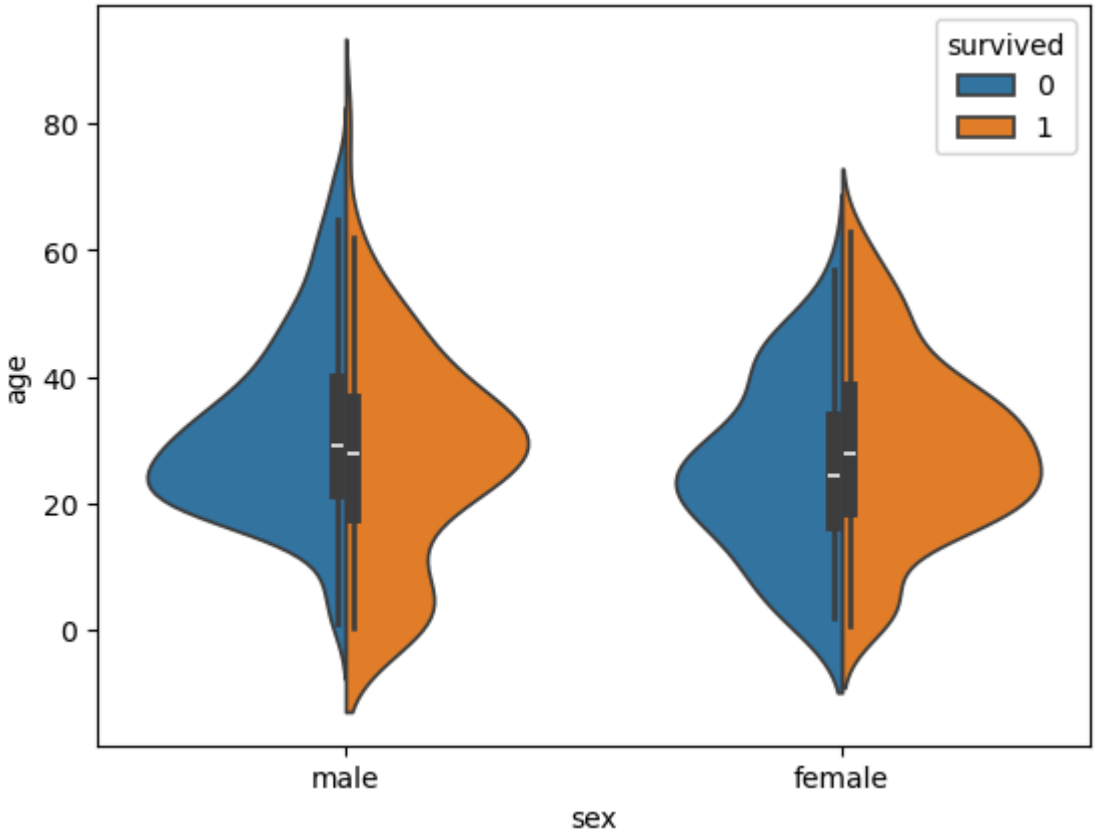
In [6]:
```
import seaborn as sns
import matplotlib.pyplot as plt

titanic = sns.load_dataset('titanic')
plt.figure(figsize=(8, 6))
sns.violinplot(x='sex', y='age', hue='survived', data=titanic, split=True

plt.title('Age Distribution by Sex on Titanic')
plt.xlabel('Sex')
plt.ylabel('Age')
plt.legend(title='Survived')
plt.show()
```



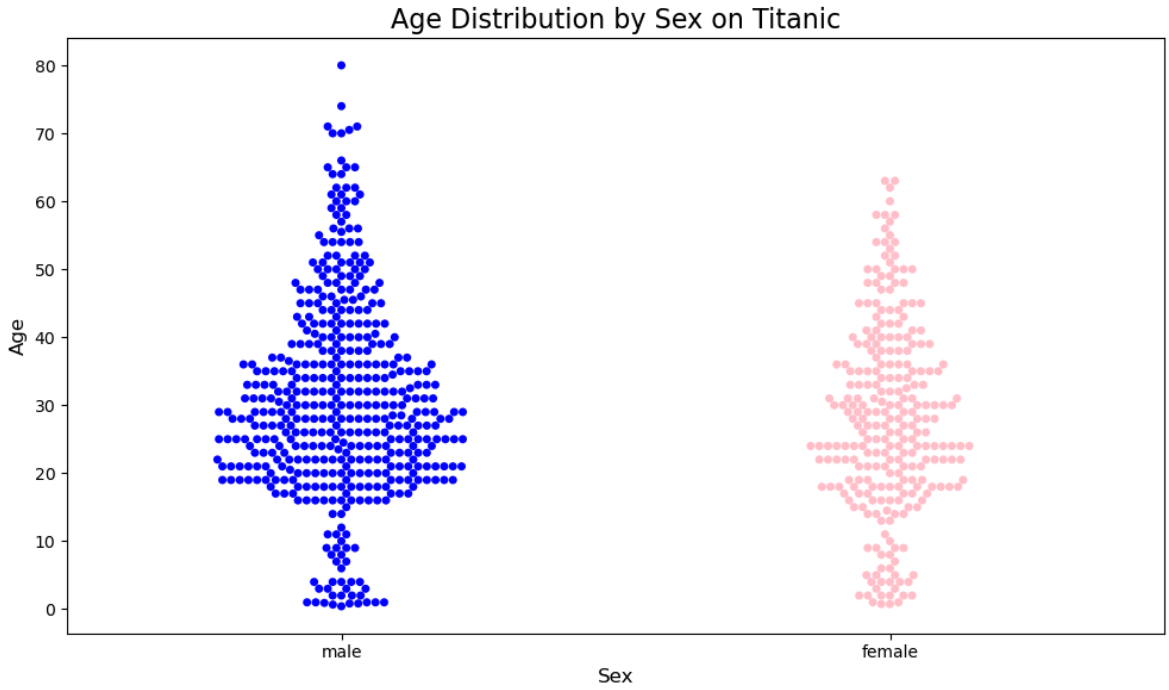
In [7]:
```
import seaborn as sns
import matplotlib.pyplot as plt
titanic = sns.load_dataset('titanic')
sns.violinplot(x='sex', y='age', data=titanic, hue='survived')
plt.show()
```



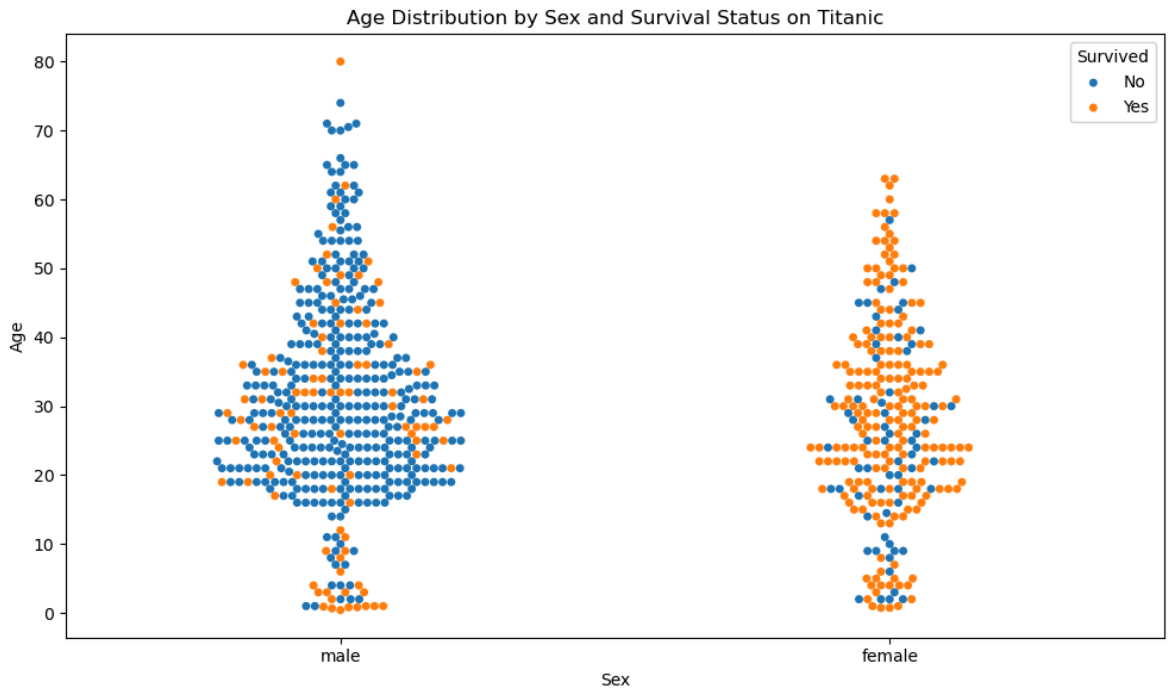
In [2]:
```
import seaborn as sns
import matplotlib.pyplot as plt
df_titanic = sns.load_dataset('titanic')
sns.violinplot(x='sex', y='age', data=df_titanic, hue='survived', split=T
plt.show()
```



In [4]:
```
import seaborn as sns
import matplotlib.pyplot as plt

df_titanic = sns.load_dataset('titanic')

plt.figure(figsize=(10, 6))
sns.swarmplot(x='sex', y='age', data=df_titanic,
              hue='sex', palette={'male': 'blue', 'female': 'pink'},
              legend=False)

plt.title('Age Distribution by Sex on Titanic', fontsize=16)
plt.xlabel('Sex', fontsize=12)
plt.ylabel('Age', fontsize=12)

plt.tight_layout()
plt.show()
```

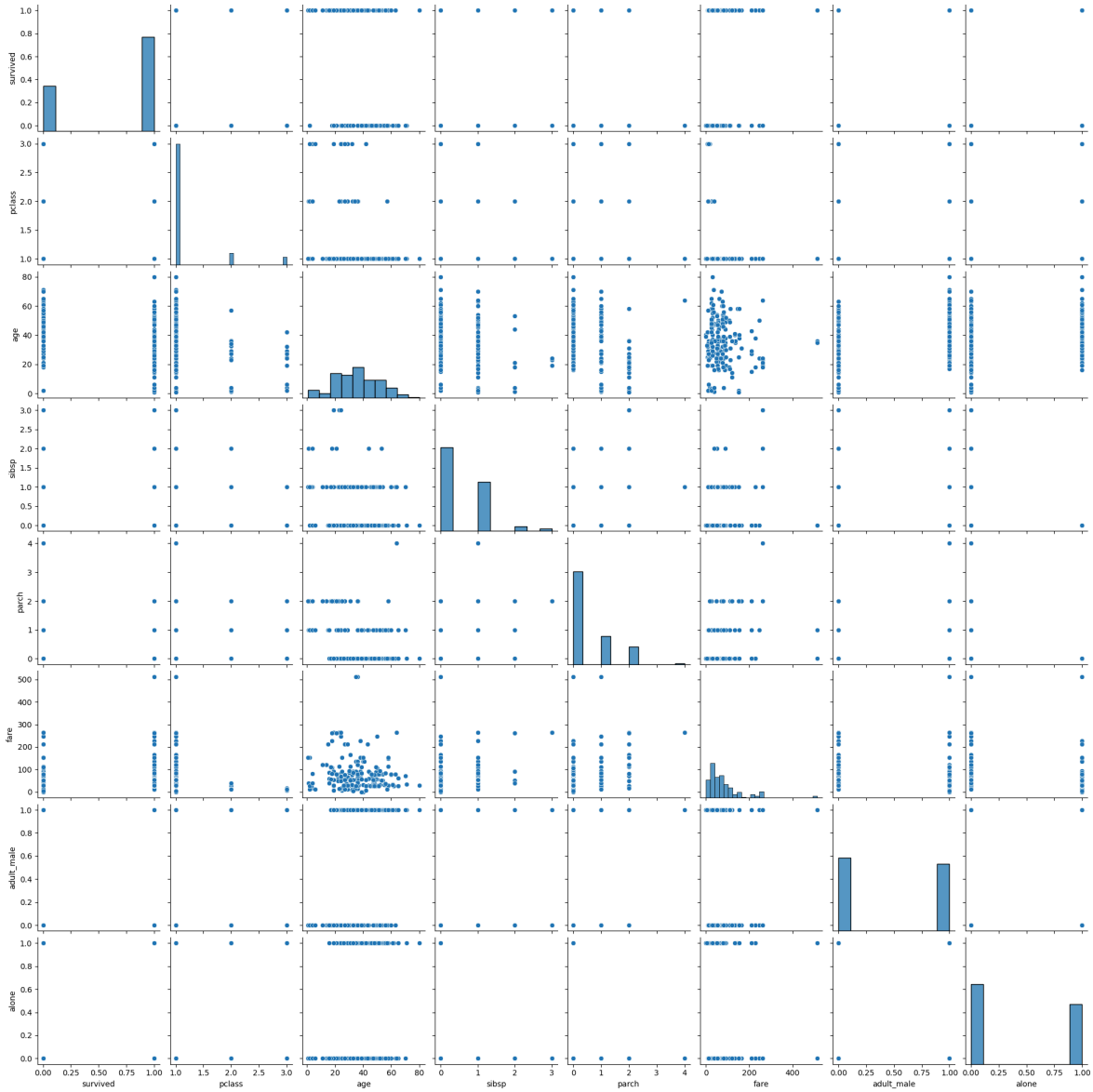


In [15]:
```
import seaborn as sns
import matplotlib.pyplot as plt

df_titanic = sns.load_dataset('titanic')
plt.figure(figsize=(10, 6))
sns.swarmplot(x='sex', y='age', data=df_titanic, hue='survived')
plt.title('Age Distribution by Sex and Survival Status on Titanic')
plt.xlabel('Sex')
plt.ylabel('Age')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.tight_layout()
plt.show()
```



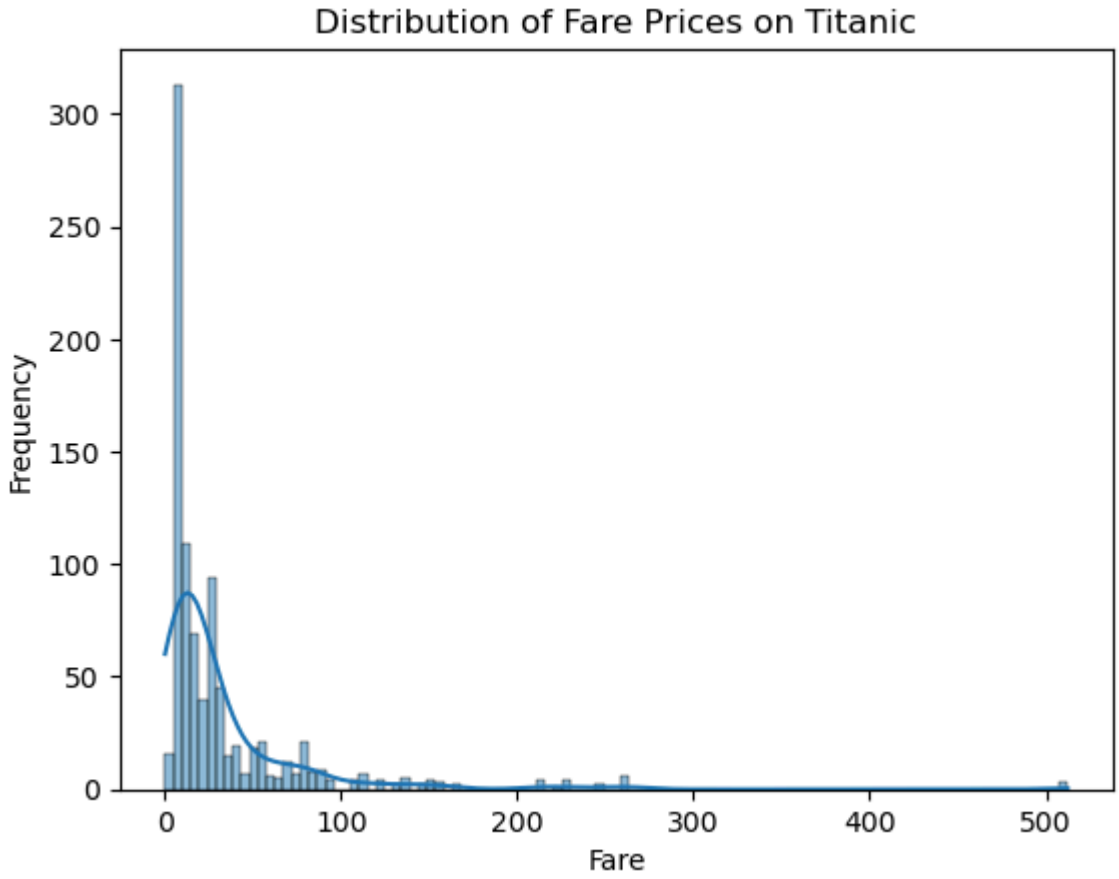
Pair plot

In [4]:
```
import seaborn as sns
import matplotlib.pyplot as plt

df_titanic = sns.load_dataset('titanic')
df_titanic_clean = df_titanic.dropna()
sns.pairplot(df_titanic_clean)
plt.show()
```

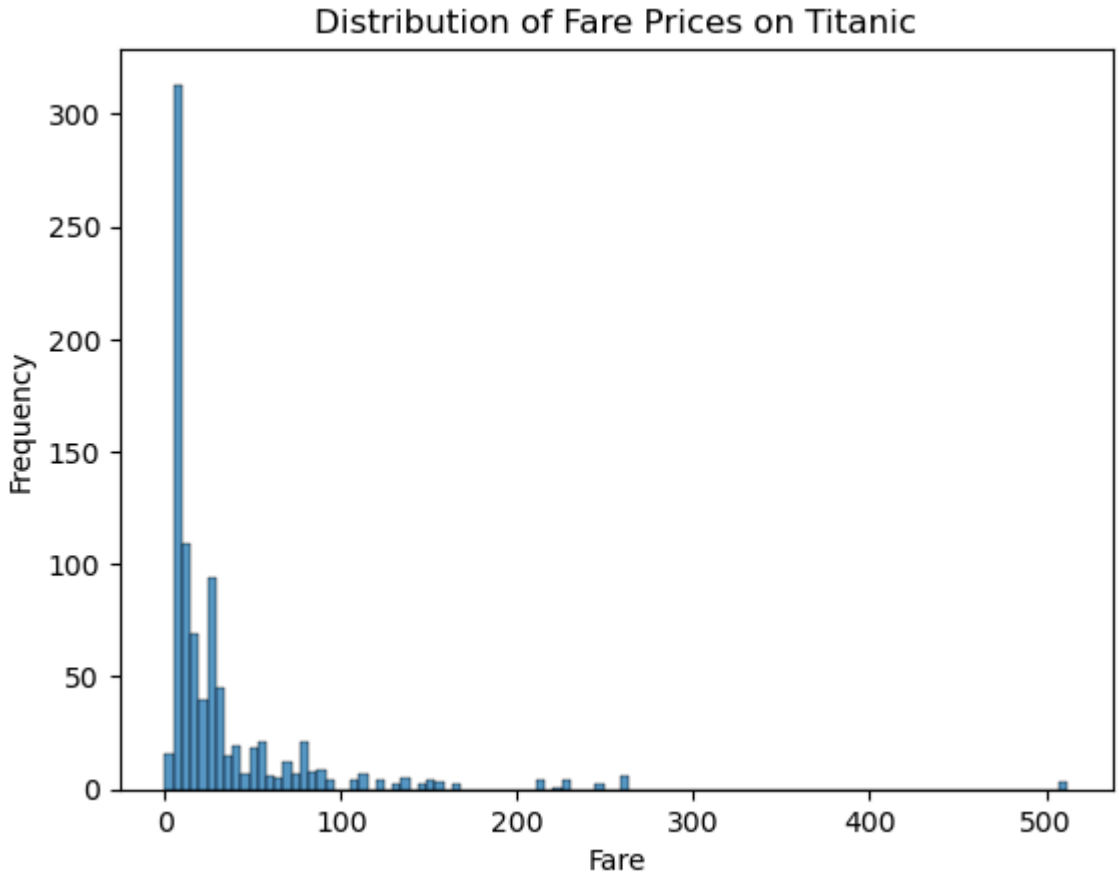


Distribution plot

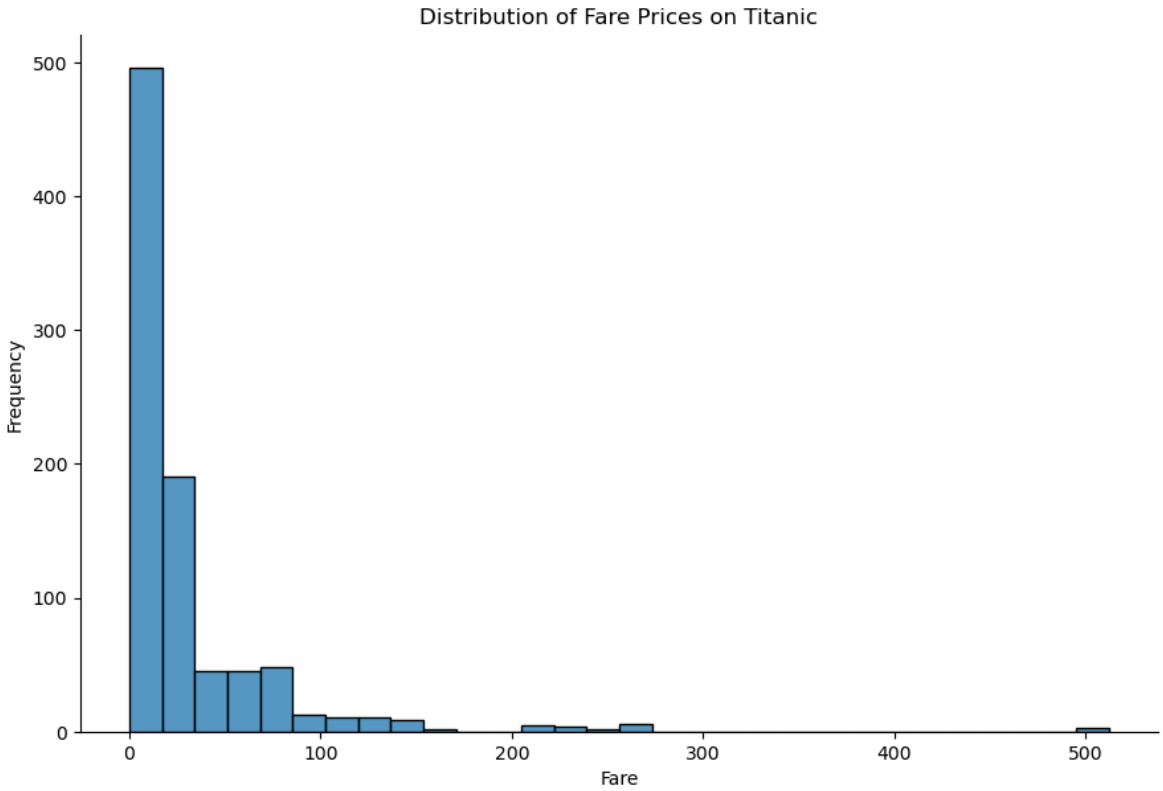
In [11]:
```
import seaborn as sns
import matplotlib.pyplot as plt

df_titanic = sns.load_dataset('titanic')

sns.histplot(df_titanic['fare'], kde=True)
plt.title('Distribution of Fare Prices on Titanic')
plt.xlabel('Fare')
plt.ylabel('Frequency')
plt.show()
```



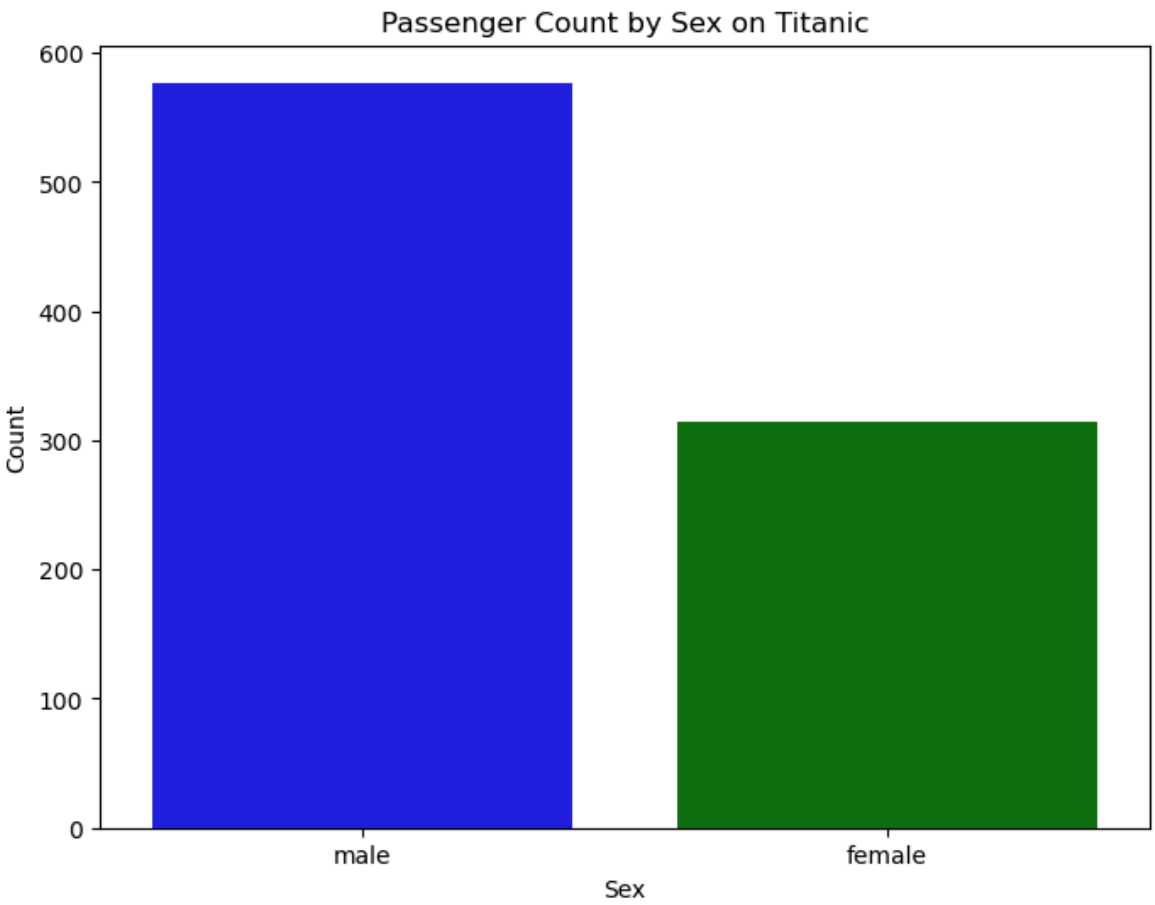
In [12]:
```
import seaborn as sns
import matplotlib.pyplot as plt

df_titanic = sns.load_dataset('titanic')

sns.histplot(df_titanic['fare'])
plt.title('Distribution of Fare Prices on Titanic')
plt.xlabel('Fare')
plt.ylabel('Frequency')
plt.show()
```



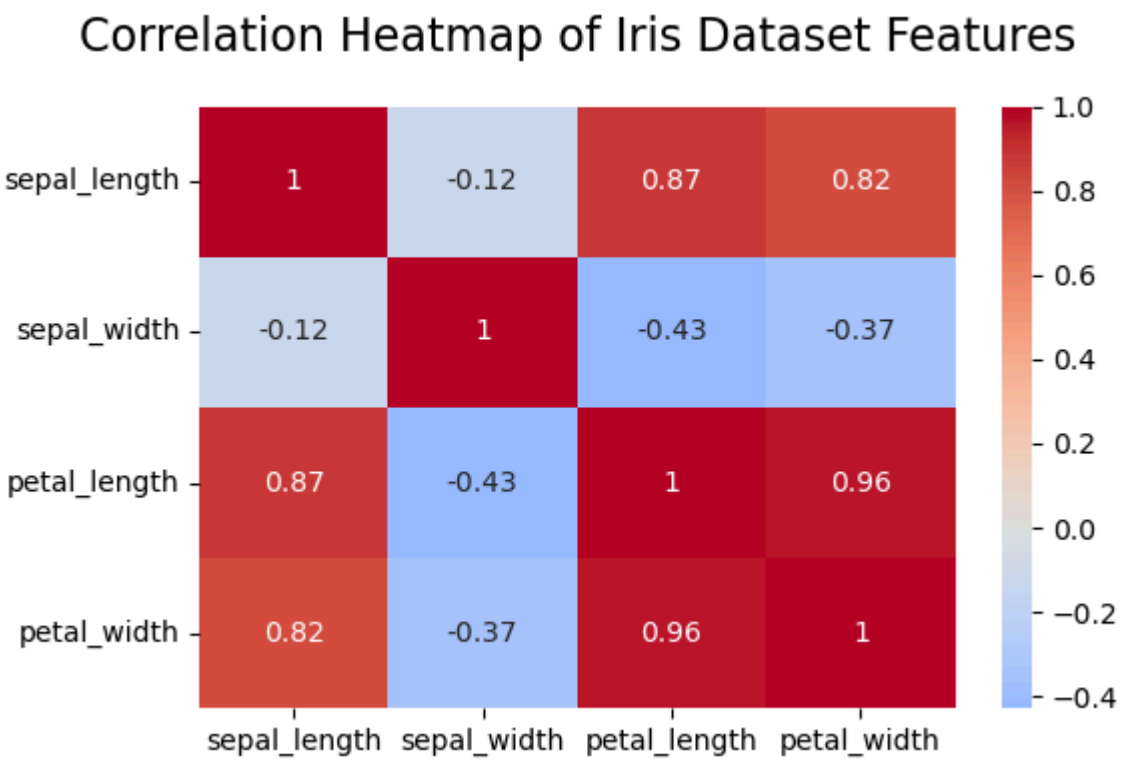
In [13]:
```
import seaborn as sns
import matplotlib.pyplot as plt

df_titanic = sns.load_dataset('titanic')

sns.displot(df_titanic['fare'], kde=False, height=6, aspect=1.5, bins=30)
plt.title('Distribution of Fare Prices on Titanic')
plt.xlabel('Fare')
plt.ylabel('Frequency')
plt.show()
```



In [14]:
```
import seaborn as sns
import matplotlib.pyplot as plt

df_titanic = sns.load_dataset('titanic')

plt.figure(figsize=(8, 6))
sns.countplot(x='sex', data=df_titanic, hue='sex', palette={'male': 'blue

plt.title('Passenger Count by Sex on Titanic')
plt.xlabel('Sex')
plt.ylabel('Count')
plt.show()
```

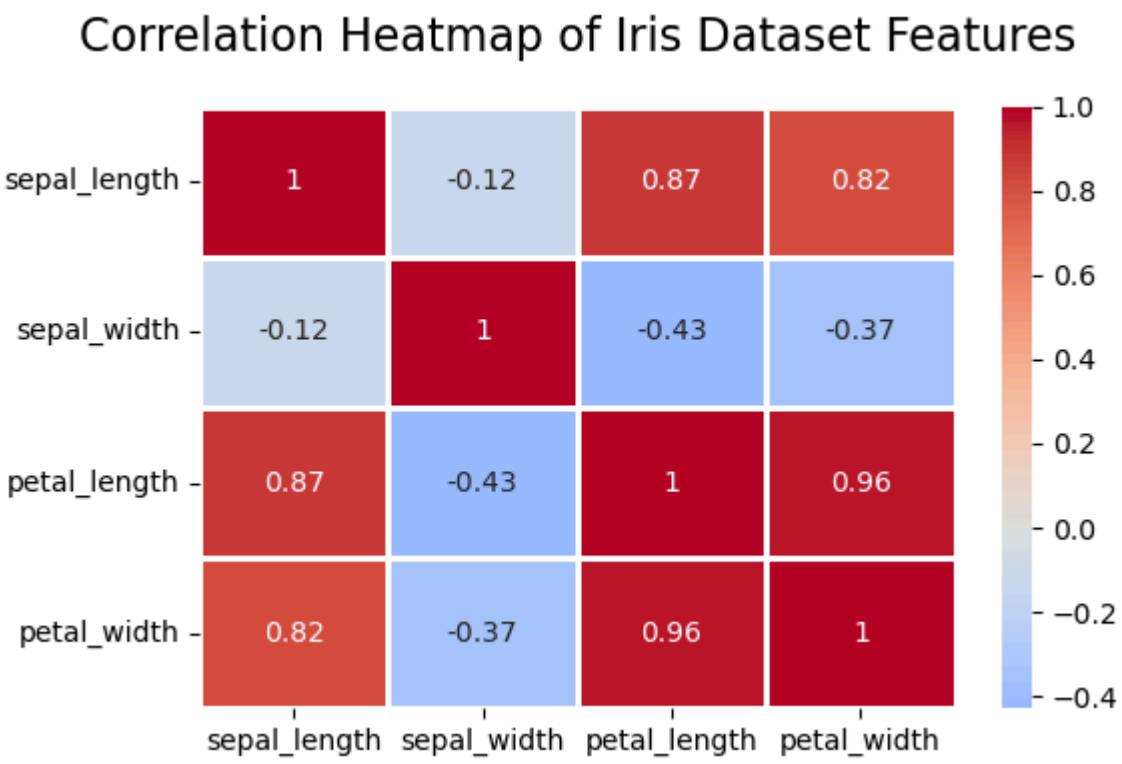


In [2]:
```
import seaborn as sns
import matplotlib.pyplot as plt

df_iris = sns.load_dataset('iris')

numerical_df = df_iris.select_dtypes(include=['float64', 'int64'])

plt.figure(figsize=(6, 4))
sns.heatmap(numerical_df.corr(), annot=True, cmap='coolwarm', center=0)
plt.title('Correlation Heatmap of Iris Dataset Features', fontsize=16, pa
plt.tight_layout()
plt.show()
```



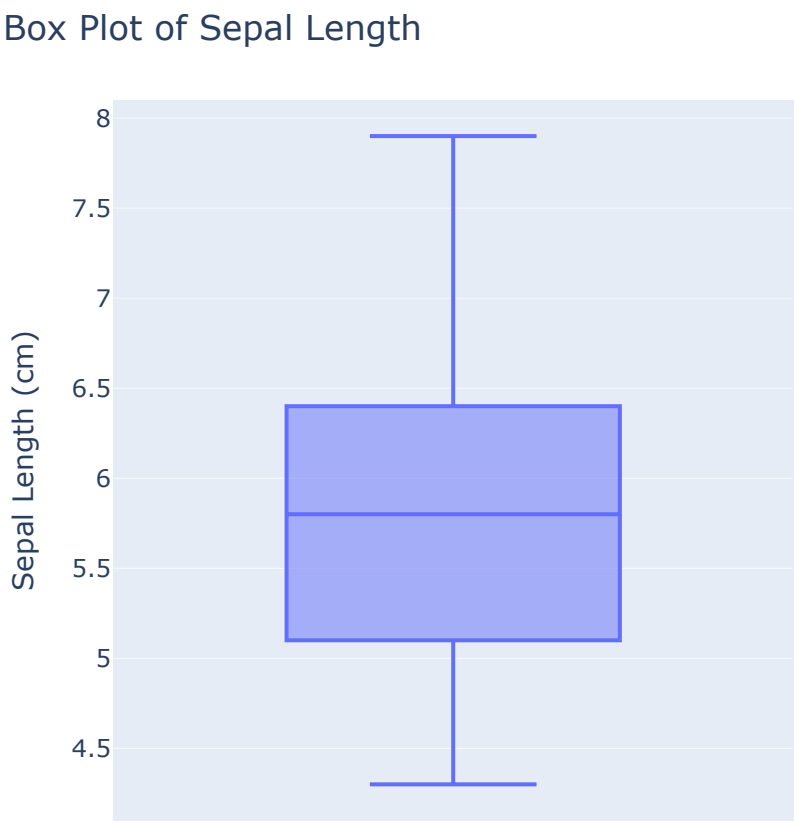
Add lines between the correlation cells

In [16]:
```
import seaborn as sns
import matplotlib.pyplot as plt

df_iris = sns.load_dataset('iris')

numerical_df = df_iris.select_dtypes(include=['float64', 'int64'])

plt.figure(figsize=(6, 4))
sns.heatmap(numerical_df.corr(), annot=True, linewidth=1, cmap='coolwarm'

plt.title('Correlation Heatmap of Iris Dataset Features', fontsize=16, pa

plt.tight_layout()
plt.show()
```

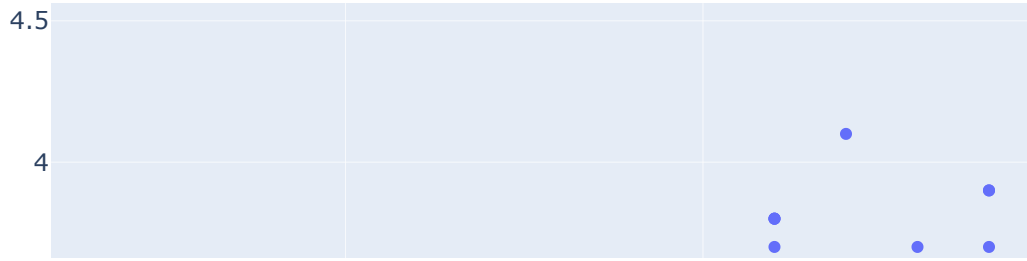


Plotly

In [17]:
```
import plotly.express as px
import seaborn as sns

# Load the Iris dataset
df_iris = sns.load_dataset('iris')

# Create box plot with correct column name
fig = px.box(df_iris, y='sepal_length')

# Update layout for better sizing
fig.update_layout(
    width=500,  # Increased from 50 to 500 for better visibility
    height=500,  # Increased from 50 to 500 for better visibility
    title='Box Plot of Sepal Length',
    yaxis_title='Sepal Length (cm)'
)

# Show the plot
fig.show()
```



In [18]:
```
import plotly.express as px
import seaborn as sns

# Load the Iris dataset
df_iris = sns.load_dataset('iris')

# Create scatter plot with correct column names
fig = px.scatter(df_iris, x='sepal_length', y='sepal_width')

# Show the plot
fig.show()
```



In [19]:
```
import plotly.graph_objects as go
import numpy as np

# Create data
x = np.linspace(-5, 5, 50)
y = np.linspace(-5, 5, 50)
X, Y = np.meshgrid(x, y)

# Create frames
frames = []
for t in np.linspace(0, 2*np.pi, 30):
    Z = np.sin(np.sqrt(X**2 + Y**2) + t)
    frames.append(go.Frame(data=[go.Surface(z=Z, x=X, y=y)], name=str(t))

# Create figure
fig = go.Figure(
    data=[go.Surface(z=np.sin(np.sqrt(X**2 + Y**2)), x=X, y=Y)],
    frames=frames
)

# Add animation controls
fig.update_layout(
    title='Animated 3D Surface Plot',
    updatemenus=[{
        "type": "buttons",
        "buttons": [
            {
                "args": [None, {"frame": {"duration": 50, "redraw": True}
```

```
                "label": "Play",
                "method": "animate"
            },
            {
                "args": [[None], {"frame": {"duration": 0, "redraw": True
                "label": "Pause",
                "method": "animate"
            }
        ]
    }]
)

fig.show()
```

Animated 3D Surface Plot

Play

Pause

In [20]:

```
import numpy as np
import matplotlib.pyplot as plt
from ipywidgets import interact, IntSlider

# Create data
x = np.linspace(0, 10, 100)

@interact(frequency=IntSlider(value=1, min=1, max=10, step=1, description
def update_plot(frequency):
    plt.figure(figsize=(10, 6))
    y = np.sin(frequency * x)
    plt.plot(x, y, 'b-', linewidth=2)
    plt.ylim(-1.5, 1.5)
    plt.title(f'Sine Wave with Frequency = {frequency}')
```

```
    plt.grid(True)
    plt.show()
```

```
interactive(children=(IntSlider(value=1, description='Frequency:', max=10,
min=1), Output()), _dom_classes=('w…
```

In [ ]:

```

```